Improve performance of existing online trackers for small object detection

UAV videos

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Problem Statement

- **Challenges** faced in detection:
  - Tiny objects $\rightarrow$ few pixels
  - Clutter, motion blur, scale variation
  - Unreliable detections
- Unreliable detections lead to:
  - Missed detections
  - Identity switches
  - Poor tracking performance

Research Gap

Existing methods are not optimized for small object detection in UAV scenarios, resulting in poor detection quality and inconsistent identity association even with tiling and ReID.

Methodology Overview

Phase 1: Detection

Understand Detectors & Trackers → Fine-tune YOLO on VisDrone → Apply Tiling (SAHI)

Phase 2: Re-Identification

Generate Crops from VisDrone → Train Siamese Network → Extend BoT-SORT with Siamese

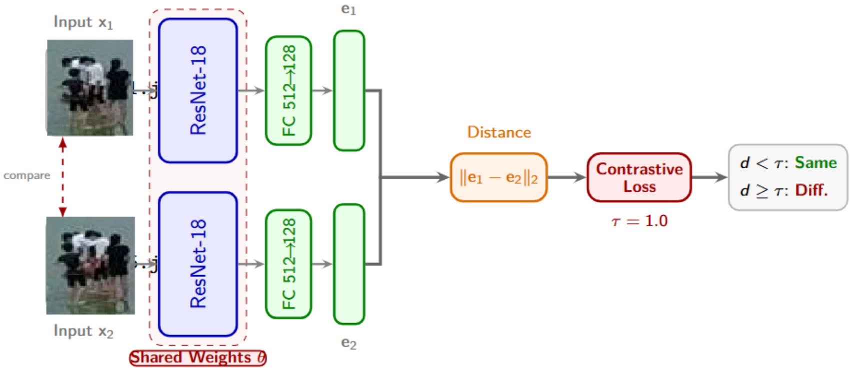
Phase 3: Full Integration

YOLOv11 + SAHI $\oplus$ BoT-SORT + Siamese ReID $\Rightarrow$ MOTA / MOTP / IDF1

Baseline Metrics

Category	Metric	YOLOv8n (Siamese NW)	YOLO11 (FT)	YOLO11 + SAHI
Tracking	MOTA	12.9%	16.73%	17.28% ↑
	IDF1	22.4%	49.02%	48.79%
	HOTA	26.0%	21.24%	–
Detection	DetA	14.2%	30.14%	–
Association	AssA	50.4%	15.57%	–
	IDP / IDR	81.7 / 13.0	–	–
Localization	MOTP	22.2%	26.51%	26.01%
Detection	Precision	81.68%	–	61.12%
	Recall	12.98%	–	49.07%
Efficiency	FPS	12.69	0.5	0.5

Siamese Network Analysis



Siamese Network Analysis

Embedding Distance

Metric	Ep 1	Ep 5
Same Obj	0.48	0.26
Diff Obj	1.15	1.04

Training Performance

Ep	Loss	Acc	Prec	Rec	F1
1	2.66	0.66	0.60	0.97	0.74
2	0.21	0.67	0.61	0.99	0.75
3	0.13	0.71	0.63	1.00	0.77
4	0.09	**0.79**	0.70	1.00	**0.82**
5	**0.08**	0.77	0.68	1.00	0.81

Future Scope

- **Full Pipeline Integration**
  YOLO11 + SAHI → Siamese-enhanced BoT-SORT → end-to-end system evaluation
- **Improve Detection & Tracking**
  Compare advanced trackers: ByteTrack, OC-SORT, NV-SORT
- **False Positive Reduction**
  NMS → Soft-NMS / Adaptive Thresholding → fewer duplicate detections
- **Enhanced Re-Identification**
  - Advanced loss functions
  - Data augmentation for small objects
- **Evaluation on Diverse Datasets**
  VisDrone → DRASHTI → improved generalization

References

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